Example 7.61

Find the local maximum and minimum of the function x^2y^2 on the line $x+y=10$.

Solution

Let the given function be written as $f(x)=x^2(10-x)^2$. Now,

$$\begin{aligned} f(x) &= x^2(100-20x+x^2)=x^4-20x^3+100x^2 \\ \text{Therefore, } f'(x) &= 4x^3-60x^2+200x=4x(x^2-15x+50) \\ f'(x) &= 4x(x^2-15x+50)=0 \Rightarrow x=0,\ 5,\ 10 \\ \text{and } f''(x) &= 12x^2-120x+200 \end{aligned}$$

The stationary numbers of $f(x)$ are $x=0,\ 5,\ 10$ at these points the values of $f''(x)$ are respectively $200, -100$ and 200. At $x=0$, it has local minimum and its value is $f(0)=0$. At $x=5$, it has local maximum and its value is $f(5)=625$. At $x=10$, it has local minimum and its value is $f(10)=0$. ■

EXERCISE 7.7

1. Find intervals of concavity and points of inflexion for the following functions:
   (i) $f(x)=x(x-4)^3$ (ii) $f(x)=\sin x+\cos x,\ 0<x<2\pi$ (iii) $f(x)=\frac{1}{2}(e^x-e^{-x})$
2. Find the local extrema for the following functions using second derivative test :
   (i) $f(x)=-3x^5+5x^3$ (ii) $f(x)=x\log x$ (iii) $f(x)=x^2e^{-2x}$
3. For the function $f(x)=4x^3+3x^2-6x+1$ find the intervals of monotonicity, local extrema, intervals of concavity and points of inflection.

7.8 Applications in Optimization

Optimization is a process of finding an extreme value (either maximum or minimum) under certain conditions.

A procedure for solving for an extremum or optimization problems.

Step 1 : Draw an appropriate figure and label the quantities relevant to the problem.

Step 2 : Find a experssion for the quantity to be maximized or minimized.

Step 3 : Using the given conditions of the problem, the quantity to be extremized .

Step 4 : Determine the interval of possible values for this variable from the conditions given in the problem.

Step 5 : Using the techniques of extremum (absolute extrimum, first derivative test or second derivative test) obtain the maximum or minimum.

Example 7.62

We have a 12 unit square piece of thin material and want to make an open box by cutting small squares from the corners of our material and folding the sides up. The question is, which cut produces the box of maximum volume?

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